

AI USE CASE INTEGRATION IN TEACHING-LEARNING

Part A: AI Use Case Profile

1. **Use Case ID:** KJS-CES-01
2. **Use Case Title:** Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning
3. **AI Application Vertical:** Climate, Energy, & Sustainability
4. **Collaborating Organization:** India Meteorological Department (IMD), Mumbai-Pune
5. **Faculty Owner:** Dr. Radhika Kotecha, Professor and Head, Department of Information Technology, K J Somaiya Institute of Technology
6. **Beneficiaries:** State Disaster Management Authorities, Municipal Corporations, Agricultural Agencies, Public Health Departments and Hospitals, Citizens, etc.
7. **Problem Statement:** Heatwaves are among the most impactful extreme weather events, with increasing frequency, intensity, and duration observed across many parts of the world. Prolonged exposure to extreme heat can adversely affect public health, agricultural productivity, ecosystems, and overall socio-economic activities. With climate change, continuous monitoring and early warning of heatwave conditions are essential for improving preparedness and reducing potential impacts. However, heatwave occurrence and severity often vary across regions due to differences in regional weather characteristics and local environmental factors, making accurate forecasting and early warning a challenging task. There is therefore a need for an effective heatwave monitoring and early warning system that can provide timely, reliable, and region-specific information to support decision-making and enhance resilience against extreme heat events.
8. **Background and Context:** Heatwaves have become an increasingly important concern in India due to rising temperatures, rapid urbanization, and changing climatic conditions. In recent years, several parts of the country have experienced prolonged periods of extreme heat, highlighting the need for effective monitoring, forecasting, and early warning mechanisms. The India Meteorological Department (IMD) continuously monitors weather conditions and issues heatwave forecasts and advisories based on established meteorological criteria.

Importantly, heatwave conditions in India are defined using region-specific temperature thresholds, considering both the normal maximum temperature and the departure from normal. For plains, where the normal maximum temperature is generally 30–40°C, a heatwave is identified when the departure from normal is 4.5–6.4°C or the maximum temperature exceeds 45°C, while a severe heatwave corresponds to a departure of more than 6.4°C or a maximum temperature above 47°C. For hilly regions, with normal maximum temperatures of approximately 20–30°C, the corresponding absolute thresholds are above 30°C for a heatwave and above 35°C for a severe heatwave, along with the same departure-from-normal criteria of 4.5–6.4°C and more than 6.4°C, respectively. For coastal regions, where normal maximum temperatures are typically 25–35°C, heatwave conditions correspond to a departure of 4.5–6.4°C or maximum temperature above 35°C, while severe heatwave conditions correspond to a departure of more than 6.4°C or maximum temperature above 37°C. These differences demonstrate that heatwave identification cannot be based on a single temperature threshold across India, making regional climatic characteristics an important consideration in heatwave analysis and forecasting.

The diverse climatic characteristics of India, together with the complex spatio-temporal nature of heatwave development, necessitate enhancing heatwave intelligence through localized monitoring, regional analysis, and advanced data-driven forecasting approaches. Recent advances in Artificial Intelligence (AI) and Internet of Things (IoT) technologies provide new opportunities to strengthen these capabilities. AI-driven spatio-temporal analysis can help identify

complex temperature patterns, regional variations, evolving hotspots, and seasonal characteristics, while IoT-enabled Automated Weather Stations (AWS) can provide continuous and localized weather observations. Their integration can enable a more comprehensive understanding of heatwave behaviour, hotspot evolution, and regional variations in extreme temperature conditions.

Along with real-time observations collected through IoT-enabled AWS, the proposed work will utilize long-term meteorological datasets provided by IMD, comprising historical temperature observations across India. These datasets contain spatial and temporal information that enables analysis of temperature variations across different climatic regions and seasons and facilitates the application of region-specific IMD heatwave and severe heatwave thresholds for identifying and analysing heatwave events. The integration of long-term historical records with localized real-time observations provides a strong foundation for studying heatwave occurrence, intensity, and evolution and for developing AI-driven heatwave intelligence, forecasting, and early-warning solutions suited to India's diverse climatic conditions.

9. Objectives:

- 1) To analyze spatio-temporal temperature patterns and heatwave behaviour through region-wise and seasonal analysis of meteorological observations.
- 2) To develop AI-driven models for region-wise maximum temperature forecasting, heatwave prediction, hotspot identification, and heatwave severity classification.
- 3) To design and deploy IoT-enabled Automated Weather Stations (AWS) for localized weather monitoring, assessment of local heat conditions, and validation of region-wise heatwave forecasts through integration of ground-based observations with meteorological data.
- 4) To develop visualization, decision-support, and early warning mechanisms for effective heatwave monitoring and dissemination of heatwave advisories.

10. Solution Approach: The proposed solution adopts a data-driven framework for heatwave prediction, monitoring, and early warning by combining historical meteorological observations from IMD with localized weather observations collected through IoT-enabled Automated Weather Stations (AWS). The framework incorporates region-wise analysis, AI-driven forecasting, localized weather monitoring, forecast validation, and stakeholder-specific advisory generation. The overall solution is organized into five interrelated phases, as depicted in Fig. 1 and described below.

Phase I: Meteorological Data Acquisition: Historical meteorological observations obtained from IMD will be collected, organized, and stored in a structured repository. Data quality assessment, pre-processing, cleaning, and transformation procedures will be performed to prepare region-wise datasets suitable for subsequent analysis and model development.

Phase II: AI-Driven Heatwave Analytics and Prediction: The processed meteorological data will be subjected to region-wise and seasonal analysis to understand spatio-temporal temperature patterns and heatwave behaviour. AI-driven models will be developed for maximum temperature forecasting, heatwave prediction, hotspot identification, and heatwave severity classification.

Phase III: IoT-Enabled Weather Monitoring Infrastructure: IoT-enabled Automated Weather Stations (AWS) will be designed and deployed to collect localized weather observations. The infrastructure will support continuous weather monitoring and provide ground-based observations for assessment of local heat conditions.

Phase IV: Forecast Validation and Heatwave Intelligence: Localized observations collected through AWS will be utilized to assess local heat conditions and validate region-wise heatwave

forecasts. Combined analysis of meteorological observations, AI-model forecasts, and ground-based measurements will facilitate development of enhanced heatwave intelligence system.

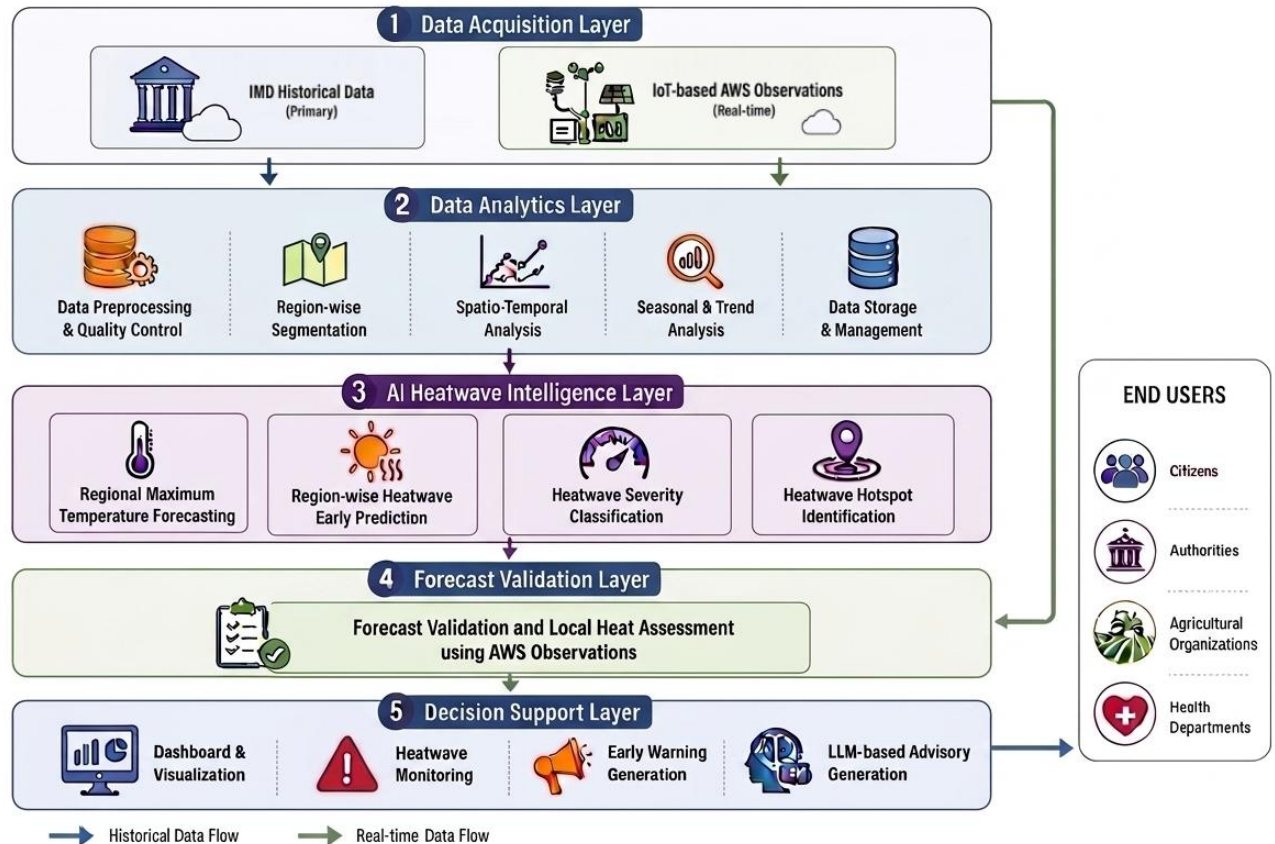


Fig. 1: Conceptual Schema

Phase V: Advisory, Early Warning, and Decision Support: A decision-support platform will be developed for visualization of forecasts, heatwave hotspots, severity levels, and monitoring information. Large Language Model (LLM)-assisted advisory generation mechanisms will be incorporated to automatically translate technical forecasts into stakeholder-specific advisories and early warnings for citizens, farmers, health agencies, and local authorities.

11. Operational Challenges:

- 1) Continuous updating and validation of AI models using newly available observations without degradation in forecasting performance.
- 2) Selection of suitable AWS deployment locations for effective localized monitoring and representative validation of region-wise heatwave forecasts.
- 3) Ensuring reliable operation of AWS infrastructure under varying environmental conditions, communication interruptions, and power constraints.
- 4) Scalability of the proposed system to support multiple regions, larger datasets, additional AWS deployments, and increasing numbers of users.

12. Dataset: <https://imdpune.gov.in/lrfindex.php>

13. High-Level Workflow:

- 1) Historical meteorological observations are acquired from IMD, Pune for analysis and model development comprising spatial and temporal attributes including Timestamp, Latitude, Longitude, and Maximum Temperature. (IT, CE, AI&DS)

- 2) Meteorological data is extracted from GRD files containing maximum observed temperature information and undergoes pre-processing, quality assessment, and preparation for spatio-temporal analysis. (IT, CE, AI&DS, CSBS)
- 3) To capture regional and seasonal variations in heatwave behaviour, the data is segmented into seven IMD-defined regions and four climatic seasons. Region-wise and seasonal analyses are performed to identify temperature patterns, heatwave characteristics, and hotspot behaviour. (IT, CE, AI&DS, CSBS)
- 4) AI-driven models are developed and utilized for maximum temperature forecasting, heatwave prediction, hotspot identification, and heatwave severity classification. (IT, Computer Engineering, AI&DS, Robotics & AI, CSBS)
- 5) In parallel, IoT-enabled Automated Weather Stations (AWS) are designed, deployed, and operated to collect localized weather observations from selected locations. (EXTC, CCE, ECE, RAI, MECH, VLSI)
- 6) Localized AWS observations are used to assess local heat conditions and validate region-wise heatwave forecasts generated by the AI models. (IT, CE, AI&DS)
- 7) Forecast outputs, monitoring information, validation results, and heatwave intelligence are integrated within a centralized decision-support platform. (IT, CE, AI&DS, CSBS)
- 8) Visualization modules present forecasts, heatwave hotspots, severity levels, and monitoring information through interactive dashboards and monitoring interfaces. (IT, CE, AI&DS, CSBS)
- 9) LLM-assisted advisory generation mechanisms convert technical forecasts and heatwave intelligence into stakeholder-specific advisories and early warnings. (IT, CE, AI&DS, CSBS, Robotics & AI)
- 10) Forecasts, alerts, and advisories are disseminated to citizens, farmers, health agencies, local authorities, and other relevant stakeholders for informed decision-making and preparedness. (IT, CE, AI&DS, CSBS)

14. Expected Outputs:

- 1) An AI-driven heatwave prediction and early warning system capable of providing region-wise heatwave forecasts and severity assessments (e.g. Fig. 2).

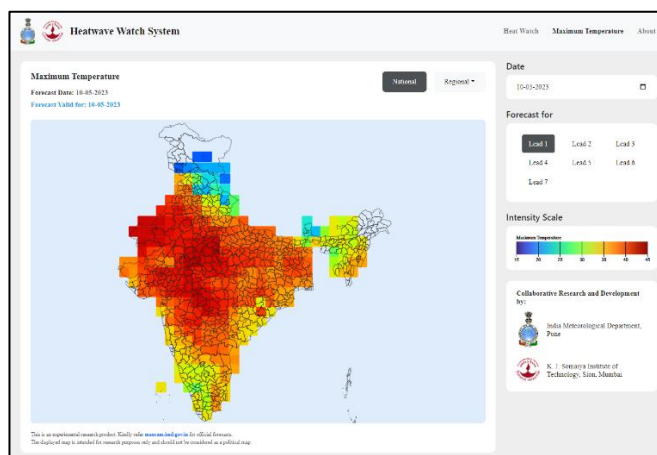


Fig. 2: Temperature Monitoring Software Screen Sample

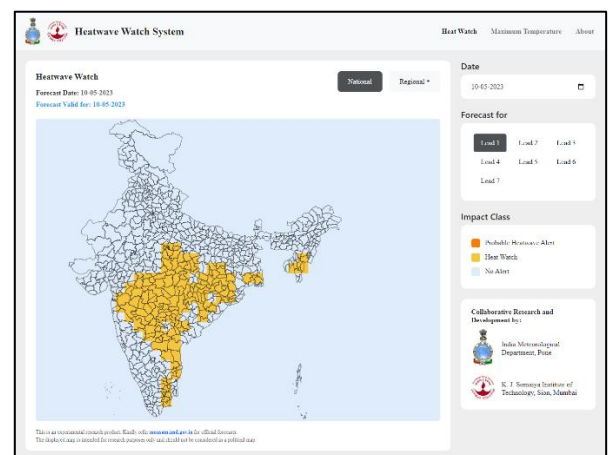


Fig. 3: Heatwave Watch Dashboard Sample

- 2) An IoT-enabled weather monitoring network comprising Automated Weather Stations (AWS) for localized monitoring and forecast validation.

- 3) A heatwave intelligence platform for visualization of temperature forecasts, heatwave hotspots, severity levels, and monitoring information (e.g. Fig. 3).
- 4) An LLM-assisted advisory generation system capable of producing stakeholder-specific heatwave advisories and early warnings.

15. Modern AI Governance and Paradigms:

Area	Relevance	Recommended Consideration
Privacy & Data Protection	Low	Ensure secure storage and controlled access to meteorological observations, AWS data, system logs, and operational information.
AI Security & Trustworthiness	High	Ensure robustness of AI models against erroneous, incomplete, or noisy data and establish mechanisms for forecast validation and performance monitoring.
Responsible AI	High	Promote transparency, fairness, and accountability in heatwave forecasting, advisory generation, and communication of uncertainty in predictions.
Generative AI / LLMs	High	Utilize LLMs for automated generation of stakeholder-specific advisories, alerts, and decision-support information while ensuring factual consistency.
Agentic AI	Low	Explore autonomous monitoring, forecast interpretation, advisory preparation, and alert dissemination workflows to support operational decision-making.
Human-in-the-Loop Decision Making	High	Enable experts, meteorologists, and authorized stakeholders to review, validate, and approve critical forecasts, advisories, and warnings before dissemination.
Regulatory / Governance Considerations	Medium	Ensure compliance with applicable guidelines related to weather information dissemination, AI governance, cybersecurity, and responsible use of forecasting systems.
